Quality Event Case Study | Discrepancy to CAPA

Synthetic walkthrough • EX-2026-018 / DCR-2026-007 / CAPA-2026-003

TRAINING EXAMPLE ONLY — Fictional organization, synthetic records, and invented personnel. Not a validated procedure, production instruction, executed approval, or assertion of regulatory compliance.

Executive scenario

A fictional biotech receiving team identifies inconsistent supplier documentation and a vendor master record. The quality reviewer must determine what is known, gather missing information, coordinate with QC, Procurement and Manufacturing, and document a potential systemic problem. The sample shows the process without implying any material has been released.

1. Workflow map

RECEIPT → SOURCE / ERP REVIEW → EXCEPTION → CROSS-FUNCTIONAL TRIAGE → IMPACT / ROOT-CAUSE ASSESSMENT → CHANGE CONTROL (IF NEEDED) → VERIFICATION → QA DISPOSITION → TREND / CAPA EFFECTIVENESS

2. Source-to-target verification

Check	Source evidence	Result	Disposition
Vendor identity	Packing slip VND-041 vs approved register VND-014	Mismatch	Open Procurement verification
Material and lot	Receipt and CoA RM-0078 / LOT-26-110A	Match	No action
Specification revision	Supplier Rev C vs controlled Rev D	Mismatch	QC/Process Owner comparison
Receiving quantity	PO 25 units / receipt 25 units	Match	No action
Disposition	Warehouse QUALITY HOLD	Matches open issue	Remain hold pending approval

3. Question routing and issue log

Issue	Owner	Request	Response / status
I-01	Procurement	Confirm supplier ID and correct approved register reference	PENDING
I-02	QC	Compare Rev C/Rev D and determine any relevant testing impact	PENDING
I-03	Manufacturing	Confirm whether proposed revised material affects intended use	PENDING
I-04	QA	Determine if controlled material change and deviation pathway are needed	PENDING

4. Example issue communication

Subject: EX-2026-018 — Vendor identifier and specification version discrepancy (RM-0078)
To: QA Operations, QC, Procurement, Manufacturing
Please review the attached source-to-system comparison. Receipt lists vendor ID VND-041 whereas the currently controlled supplier register displays VND-014; supplier documentation also references specification Rev C instead of controlled Rev D. The material remains in QUALITY HOLD. Procurement: please confirm the approved vendor record. QC/Manufacturing: please assess the specification revision and intended-use impact. Record decisions and linked evidence in the exception log; do not modify approved source documents without controlled correction.

5. Investigation structure

- Problem statement: identify exact records, affected material and timestamp; avoid asserting product impact before evidence.
- Immediate controls: maintain applicable hold and restrict unauthorized use while QA triages impact.
- Evidence plan: supplier approval history, material/spec revision records, receipt data, data-entry audit trail and change request.
- Potential causes to test: incorrect vendor code mapping; stale specification attachment; lack of receipt pre-check; unclear revision handoff. These are hypotheses, not findings.
- Risk review: confirm whether other receipts, vendors, materials, or open orders have the same mapping pattern.

6. Proposed corrective and preventive actions

Action	Type	Accountable role	Verification evidence
Reconcile vendor ID and preserve original transaction history	Correction	Master Data / Procurement	Authorized change record and audit trail
Review Rev C / Rev D impact and decide material disposition	Correction / impact	QC / QA	Approved assessment and disposition
Add vendor-ID verification step to incoming QC work instruction	Preventive	Process Owner / QA	Approved DCR and controlled revision
Train impacted receiving and QA roles on revised step	Preventive	Training Coordinator	Training assignment and completion evidence
Review a defined subsequent sample of receipts for recurring mismatch	Effectiveness	QA	Sampling plan and documented results

7. Effectiveness check — planned, not performed

Proposed indicator: 0 unexplained vendor-ID mismatches in the next 30 qualified receipts over a defined 60-day observation period, with 100% of exceptions recorded and owned. These are sample design targets, not measured outcomes. If insufficient receipts occur, QA must document an adjusted justified sample and continue monitoring.

8. Closure gate

Case remains OPEN in this demonstration. Closure would require documented approvals, confirmed source-to-system correction, material disposition by an authorized party, completed actions, and a QA-approved effectiveness conclusion. No signatures or release decisions are fabricated.

9. Technical-writing choices

- Task-oriented instructions instead of policy-only language.
- Distinct verified facts, open issues, hypotheses, actions and acceptance criteria.
- Readable field comparison designed for QC/Procurement communication.
- Explicit actor/responsibility at each handoff.
- Versioned sample documents and transparent fictional status.

REGULATORY / TECHNICAL REVIEW ADDENDUM — November 2026

Independent technical writing portfolio enhancement. Additional validation gates and examples are illustrative, not authorized production instructions. Regulated applicability varies by product, process, record and organization.

Investigation method and alternative explanations

Problem statement: an incoming raw-material lot shows conflicting supplier identity and specification revision across a packing slip, vendor master and certificate. The investigation asks whether the root cause arose from source documentation, an ERP mapping table, an obsolete approved supplier record, or human transcription.

Collect independent evidence before concluding root cause: timestamped vendor-master history, PO and revision history, receipt transaction, certificate document, document-control register, user activity history where appropriate, and cross-functional interviews. Separate confirmed facts, hypotheses, and unresolved questions.

Quality impact is assessed before proposed closure, including whether material was used, whether associated lots or records are affected, and whether containment is sufficient. If a drug component lot is implicated, retain quality-unit disposition authority and applicable 21 CFR 211.84 controls.

CAPA lifecycle and effectiveness

Containment is immediate protection (place material on hold; notify the authorized quality unit). Correction addresses the immediate wrong record without erasing history. Corrective action addresses a supported root cause (for example, controlled supplier-mapping review). Preventive action strengthens the system where justified; not every event requires both separately.

Example effectiveness plan: after approved system and procedure changes, QA samples 20 consecutive applicable receipts over a defined observation interval and checks supplier identity and specification revision. Acceptance is zero recurrence of the same verified root-cause mismatch; a recurrence reopens the assessment. The sample size and threshold are proposed company rules, not FDA universal numbers.

No CAPA is described as effective until evidence is reviewed by authorized quality personnel; open actions and planned verifications remain plainly marked OPEN/PROPOSED in the sample records.

Training impact and review traceability

Before activation, quality/document owners identify which users need new vendor-verification instructions and whether competency evaluation is required. Maintain an effective revision ID, date, audience and completion evidence, not fictional initials claiming sign-off.

Link event EX-2026-018 to DCR-2026-007, CAPA-2026-003, the corresponding approved controlled documents and the records of actual verification where they exist. The training example deliberately has no executed signatures.

Document-specific reference basis and status

FDA 21 CFR 211.22 — quality unit authority (full URL in portfolio review report)

FDA 21 CFR 211.84 — component sampling/testing/release (full URL in portfolio review report)

FDA 21 CFR Part 11 — electronic records and e-signatures (full URL in portfolio review report)

FDA Part 11 scope and application guidance (full URL in portfolio review report)

FDA Data Integrity and Compliance With Drug CGMP (2018) (full URL in portfolio review report)

ICH Q9(R1) Quality Risk Management (2023) (full URL in portfolio review report)

ICH Q10 Pharmaceutical Quality System (full URL in portfolio review report)

ICH Q8/Q9/Q10 Q&A R5 (May 2026) (full URL in portfolio review report)

Status: technical/editorial review against publicly available references completed at a portfolio-example level; no licensed-standard full-text conformance audit, live environment testing, site QA approval or legal certification performed.

Appendix: November 2026 standards and documentation applicability review

PUBLIC SYNTHETIC PORTFOLIO SAMPLE — This appendix records source-checking and editorial scope boundaries. It does not make the fictional organization regulated, certify a system, or replace actual company policies and competent quality/legal review.

Pharmaceutical regulatory scope and data-integrity editorial verification

- 21 CFR 211.22: fictional QA unit responsibility is distinct from QC testing and procurement approval; incoming component disposition must not be inferred from data-field reconciliation alone.
- 21 CFR 211.84: describe quarantine, identity testing and QC release expectations only to the extent that the hypothetical material falls under the rule; signed review of a supplier certificate is not automatic approval.
- 21 CFR 211.180/211.192 and FDA data-integrity guidance: reconcile original records, trace corrections, record investigations, preserve legibility and demonstrate review of unexplained discrepancies.
- 21 CFR Part 11: evaluate record and electronic-signature predicate-rule scope; do not claim that every spreadsheet or SOP falls under Part 11.
- ICH Q9(R1)/Q10 and FDA May 2026 Q8/Q9/Q10 Q&A: risk-based investigation, change control, oversight, effectiveness checks and management review are guidance/model concepts, not blanket product approvals.
- 2026 FDA QMSR/21 CFR 820 and February 2026 Computer Software Assurance: relevant to in-scope medical-device workflows; not universally required for this hypothetical drug-component process.
- Document design check: explicit document ID/version/status, role-based responsibilities, objective acceptance criteria, accurate dates, no invented e-signatures or final QA approvals, and clear synthetic label.

Reference links and recheck notes

https://www.ecfr.gov/current/title-21/chapter-I/subchapter-C/part-211

https://www.ecfr.gov/current/title-21/chapter-I/subchapter-A/part-11

https://www.fda.gov/regulatory-information/search-fda-guidance-documents/data-integrity-and-compliance-drug-cgmp-questions-and-answers-guidance-industry

https://www.fda.gov/medical-devices/postmarket-requirements-devices/quality-management-system-regulation-qmsr

Verified against published authoritative source pages as of November 2, 2026. No proprietary standards were audited clause-by-clause. Actual application must be assessed based on product, system use, region, and record type. Proposed procedures have not been executed in a production environment.

Independent second-pass document review | November 2, 2026

REVIEWED PORTFOLIO DEMONSTRATION — this appendix records a document-specific editorial and source-scope check. The main procedure and forms are fictional and not approved for production, executed testing or regulated use. Original source versions are retained as editable files in this portfolio.

A. Document-specific findings and checks

Review objective	2026 reviewer correction / expectation	Verification evidence
Evidence chain	Capture mismatch discovery, affected lot, version metadata, receiving/approved supplier data, status hold and investigation owner.	Each allegation identifies synthetic evidence and distinguishes allegation from confirmed fact.
CAPA decision	Separate immediate correction from systemic corrective/preventive actions; root-cause hypothesis requires validation.	Document action due date, accountable owner, verification test and effectiveness acceptance criteria.
Release decision	No CAPA closure or manual spreadsheet correction automatically removes a material hold.	Show independent quality-unit approval and material-specific release evidence, if applicable.

B. Applicable regulatory / guidance boundaries

Review objective	2026 reviewer correction / expectation	Verification evidence
Drug CGMP / Quality unit	21 CFR 211.22, 211.80, 211.84, 211.100, 211.160 as relevant	Mandatory only when the fictional activities actually fall within covered finished-drug operations; paperwork precheck is not QC material disposition.
Records integrity / electronic signature	21 CFR 211.180, 211.188, 211.194; Part 11 conditional	Trace data to author, time, source and approved version; do not assume that every spreadsheet is a Part 11 record.
Risk and change	ICH Q9(R1), ICH Q10; FDA 2018 final data integrity guidance	Nonbinding guidance/system model as adopted or implemented; distinguish a risk-based recommendation from regulation.

C. Evidence and sign-off boundaries

Status: Editorial second-pass reviewed; technical procedure not executed; company-specific SOP approval not obtained.

All tests, screenshots, users, suppliers, dates, IDs, incident observations, performance metrics and signatures in the demonstration are synthetic.

Next practitioner action: confirm source applicability for the actual jurisdiction and product; conduct SME/user testing and accessibility review; update controlled revision and approval fields.

No color, watermark, revision number, PDF export or reference link by itself demonstrates regulatory compliance.

D. Authoritative primary-source references

FDA drug CGMP 21 CFR Part 211: https://www.ecfr.gov/current/title-21/chapter-I/subchapter-C/part-211

FDA 21 CFR Part 11 scope: https://www.fda.gov/regulatory-information/search-fda-guidance-documents/part-11-electronic-records-electronic-signatures-scope-and-application

FDA drug CGMP final data integrity: https://www.fda.gov/regulatory-information/search-fda-guidance-documents/data-integrity-and-compliance-drug-cgmp-questions-and-answers

ICH Q9(R1)/Q10: https://admin.ich.org/page/quality-guidelines

FDA device QMSR Feb 2026: https://www.fda.gov/medical-devices/postmarket-requirements-devices/quality-management-system-regulation-qmsr